USN | | | | | | | | | | [22CSBS207]

THIRD SEMESTER B.Tech EXAMINATION, DECEMBER -2023

COMPUTATIONAL STATISTICS

Time: 3 Hours **Maximum Marks: 100**

Instructions: i. Missing data may be suitably assumed.

ii. Answer All Questions

		PART-A			**05 X 02=10**
1.	**a.**	What are the methods of estimating parameters?	1	CO1	**02**
	b.	Discuss the need of discriminant analysis.	1	CO 2	**02**
	c.	Which plotting techniques is used to preserve relationships between dissimilarity of variables and how?	1	CO 3	**02**
	d.	What is a factor transformation matrix?	1	CO 4	**02**
	e.	List out the types of Clustering Algorithm?	1	CO 5	**02**
		PART-B			**06 X 05=30**
2.	**a.**	Find P(X<0)? if X~N(−5,4)	3	CO 1	**05**
		OR			
	b.	Which two parameters are there in multivariate normal distribution and give the formula for multivariate normal distribution?	2	CO 1	**05**
3.	**a.**	What are the ways of factor scores that can be Estimated and explain any one method?	2	CO 2	**05**
		OR			
	b.	List out advantages and disadvantages of (LDA)?	2	CO 2	**05**
4.	**a.**	What are the different Types of Histogram	2	CO 3	**05**
		OR			
	b.	Draw the histogram for the lifetime of 400 neon lamps below data.	3	CO 3	**05**

Lifetime (in hours)	300 – 400	400 – 500	500 – 600	600 – 700	700 – 800	800 – 900	900 – 1000
Number of lamps	14	56	60	86	74	62	48

5.	**a.**	How do you determine the number of factors in factor analysis?	2	CO 4	**05**
		OR			
	b.	What are the ways to calculate factor scores?	2	CO 4	**05**
6.	**a.**	Explain about Cluster Analysis.	2	CO 5	**05**
		OR			
	b.	List out the Different types of clustering techniques.	2	CO 5	**05**

7.	a.	Define a multivariate normal density function, of order n?	2	CO 1	05
		OR			
	b.	Among 900 people in a state 90 are found to be chapati eaters construct 99% confidence interval for the true population?	2	CO 5	05
		PART-C	**06 X 10=60**		
8.	a.	Explain Multivariate Normal Distributions functions?	2	CO 1	10
		OR			
	b.	Explain Conditional Mean, Conditional Variance, Conditional Covariance in multivariate conditional distribution?	2	CO 1	10
9.	a.	Explain the procedure involved in Discriminant analysis?	2	CO 2	10
		OR			
	b.	Discuss the Linear Discriminant Analysis (LDA) Functions and Limitations of LDA?	2	CO 2	10
10.	a.	Write the steps involved in performing Principle Component Analysis.	2	CO 3	10
		OR			
	b.	What are the ways to retain Principle Component in Principal Components Analysis (PCA)?	2	CO 3	10
11.	a.	Explain any two methods in estimating factor scores.	2	CO 4	10
		OR			
	b.	Explain the Factor Rotations and its methods with example?	2	CO 4	10
12.	a.	Explain Agglomerative, Divisive hierarchical clustering and dendrogram with suitable example?	2	CO 5	10
		OR			
	b.	Explain the Steps involved in the K-means Clustering Algorithm.	2	CO 5	10
13.	a.	What are the different types of distance measurement and give formula for any one method?	3	CO 1	10
		OR			
	b.	Cluster the following eight points (with (x, y) representing locations) into three clusters: A1(2, 10), A2(2, 5), A3(8, 4), A4(5, 8), A5(7, 5), A6(6, 4), A7(1, 2), A8(4, 9).	3	CO 5	10